



Document Management System



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Ministry of Health
The Royal Hospital
Department of Fire Safety

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Title: Fire Code Policy

POLICY FOR FIRE CODE RESONSE

POLICY

The Royal Hospital shall have in place at all times a coded system to be initiated in the event of any confirmed fire incident. This shall be known as the **FIRE CODE** . The existing Royal Hospital fire emergency telephone extension number **9777** shall remain and serve as the Code Red number.

OBJECTIVE

The objective is to provide on a permanent 24 hour basis a selected **“On Duty”** team or managers or otherwise competent persons to respond in the event of a confirmed outbreak of fire. The team shall be known as the **FIRE CODE TEAM** and shall respond immediately to the location of an occurrence of fire in order to bring authority and take control of the incident.

1. FIRE CODE TEAM MEMBERS

1.1 The fire Code Team shall be as follows:

During Normal Working Hours:

D.G or whoever act .

Fire Officer.

Nursing Officer.

Security.

PRO.

Outside normal working hours:

Duty Director.

Fire Officer.

Eng Controller.

Duty Nursing Officer.

PRO.

Security.

Accident & Emergency Dept.

1.2 Outside of normal working hours the fire code team shall respond to and assess the severity of an incident.

1.3 The fire code team having assessed the severity of the incident shall make the decision whether or not to place a call to the **Civil Defense Fire & Rescue Services**.

1.4 The decision to call the **Civil Defense Fire & Rescue Services** shall be based solely upon whether or not the fire is in progress and if it is within or beyond the control of staff on duty using the life safety equipment provided.

1.5 If required, the fire code team shall initiate a recall of key staff (Key staff to be defined).

1.6 In such circumstances where an evacuation is required, the fire code team shall supervise a progressive evacuation of patients.

1.7 The fire code team shall terminate any incident based on their decision after arrival and assessment of the incident.

2. ARRANGEMENTS FOR ALERTING THE FIRE CODE TEAM MEMBERS.

2.1 Information that a fire incident may *possibly* be in progress shall be received as an electronic signal by the hospital switchboard operator at the fire alarm panel, and / or by the maintenance operator via the control room.

2.2 Information confirming that a fire is actually in progress shall be received by the staff via the fire Code **9777**. Upon receipt of such information the staff shall record the location, the name of the caller and details of the fire.

2.3 Upon confirmation of any fire incident in progress the staff shall immediately alert the fire code team members by CPN phone or Landline phone announcing the location of the incident.

2.4 The announcement shall be made calmly and clearly in English and Arabic and shall be repeated twice.

2.5 Once the announcement has been made the staff shall remain vigilant in order to receive or dispatch further information.

3. DEBRIEFING

3.1 A debriefing shall be made at the termination of any fire incident which requires fire code team response.

4. DOCUMENTING THE INCIDENT

4.1 An incident report shall be completed and a written report shall be submitted by the fire officer as soon as possible following any fire code team response.

5. EVALUATING THE FIRE CODE VIA A SIMULATED RESPONSE DRILL

- 5.1 The fire code number shall be test called once per Month. All team members shall be alerted via their CPN phone.
- 5.2 The announcement shall be made clearly in English and Arabic and shall be repeated twice.
- 5.3 Upon receipt of the fire code drill information the fire code team members shall respond immediately to the location.
- 5.4 A written record of the team members in attendance at the location shall be made.
- 5.9 The fire code drill shall be debriefed and any flaws shall be noted and improved upon.

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Ministry of Health
Royal Hospital



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Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Fire Safety

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Introduction

Royal Hospital aims to minimize the loss of life and personal injury through fire/accident and associated hazards. When the fire occurs, there will be volumes of smoke, most of which will be toxic. To ensure that patients, visitors and staff have the best chance of survival it is essential to implement evacuation procedures whenever fire occurs in an area.

The intention of this guidance is to present a series of easy to reference guidance documents presented so that each section can be viewed separately, though they are concerned with the same subject, Fire and Safety Management, policy and procedures to assist in the development and implementation of Fire and Safety Management.

Royal Hospital Buildings requires the development of operational procedures to:-

1. Well designed buildings with unobstructed escape routes that enable the occupants of the building to move quickly around the buildings to places of safety without the risk of fire.
2. Prevent fires occurring in all buildings owned or occupied by the Royal Hospital.
3. Minimize the loss of life and personal injury through fire and associated hazards.
4. Ensure that when a fire occurs there:-
 - a. Will be trained staff able to react quickly in a calm and professional manner, evacuating patients to predetermined locations.
 - b. Are sufficient personnel and equipment available whenever the building is occupied for evacuation purposes.
 - c. Trained staff capable of fighting the fire.
 - d. Good communications systems providing information about the fire.
1. Procedures in place to quickly reinstate healthcare facilities and services following a fire.

To achieve these aims there needs to be in all premises owned occupied or managed by the Royal Hospital:-

1. Local fire policies (compliant with national policies) developed to ensure that the threat from fire is countered and ensure maintenance of healthcare provision following a fire.
2. Competent people (all employees/contract company staff), who are regularly trained and practice in incident prevention and incident response procedures, including emergency evacuation techniques, firefighting equipment use.

4. Training programmes that ensure staff and contract staff are competent with regard to fire prevention and incident response.
5. Evacuation drills and exercises that will be conducted in each department at intervals of not less than once a year.
6. A programme to reduce the effects of arson.
7. A policy ensuring that the needs for security do not prevent the speedy evacuation of occupants from buildings.
8. Safe systems of work.
9. Safety measures including fire warning system and firefighting equipment.

Management Planning

The content of this document should be used as the model on which to base local policies. In addition there is a generic document outlining all of the factors to be included in a comprehensive set of policies and procedures including a carefully prepared program for dealing with fire prevention, fire – fighting and the movement or evacuation of patients and other building occupants in an emergency. The program should include the implementation of precautionary measures to prevent the occurrence of fire, and provisions for dealing with outbreaks (including minimizing the impact of fire).

Policies should also include instruction and training to ensure that every member of staff has a clear understanding of their role, and will cooperate in taking effective emergency action. Although emphasis should be on fire prevention, an essential ingredient of any fire safety policy is the preparation of an operational strategy for immediate implementation when a fire emergency arises. This strategy should set out the emergency procedures and should be prepared to suit the circumstances of individual premises and departments. All staff should be familiar with the content of this strategy, as it is one of the most important features of a fire policy. The fire safety coordinator has overall responsibility for drawing up and maintaining comprehensive fire precautions, fire safety policies and programs of improvement, but managers at each appropriate level in the organization should also be involved. 24 hours a day there must be somebody responsible (fire officer) for and take charge of the fire action plans and procedures in the event of fire.

The Civil Defence will have a variety of procedures available for dealing with fires in building. These procedures are aimed, primarily, at ensuring the right resources are in the right place at the right time throughout the Civil Defence's area. Royal Hospital management must ensure that all their policies and procedures are fully discussed with the Civil Defence and area compatible with the Civil Defence procedures.

The development of a Fire Action Plan divided into 3 areas will ensure that all aspects of fire are discussed and measures put in place to reduce the risk of fire to an acceptable level. The 3 parts of the plan are:-

1. The Fire Action Plan
2. The Training Plan
3. The Maintenance Plan

Fire Action Plan

The first requirement of the Fire Plan is to do all that is possible to prevent fire from occurring by introduction of policy and procedures to:-

- Stop all noncompatible storage e.g. highly flammable liquids and combustibles;
- Ensure corridor and exits are not blocked;
- Plan and coordinate for adequate storage areas;
- Control of hazards in the workplace;
- No alterations without the necessary approvals;
- Consultation with the fire Supervisor on management proposals

Policy and procedures are required to inform staff of the correct procedures to follow in the event of fire. Policy and Procedures must be controlled by administration and readily available to staff in a suitable format that is controlled to ensure that only up to date information is contained within the documents that are reviewed regularly to ensure that changes within the building are included in revisions. Every time new equipment is introduced or the building altered the evacuation plan for that area must be reviewed.

Communications

It is very important that there is effective communication of all information regarding fire and safety management. Communications is not just about policy and procedure documents that are up to date and easily available to staff; but, also such things as training and the fire warning system that communicates to staff where the fire is so that actions can take place immediately without the need for orders to be given from managers.

Fire Evacuation

There are many areas of the Royal Hospital that can evacuate to outside (administration, out patients departments) most patients are too ill to be moved far and the movement will adversely affect the health of patients. Therefore, the evacuation principle used is called Progressive Horizontal Evacuation (PHE) where patients are moved to the next nearest ward so that treatment and care can continue.

Staff will receive training in how to evacuate patients using a variety of rescue techniques and in some cases specialist equipment by the Fire Safety Supervisor. Ward/departmental managers and Supervisors to understand the principles of means of escape and for hospitals stage 1 and stage 2 horizontal evacuation.

Training Plan

Once all the items mentioned above forming the fire plan and the various procedures are agreed a mandatory attendance annual training programme should be developed for all staff. Staff need to have an understanding of fire risks and know what to do in the event of fire so that fire safety procedures can be applied effectively. This applies to all staff without exception. Senior management and senior medical staff should lead by example.

The Fire Safety Coordinator should be responsible for ensuring a training plan is developed and programmed. This programme should reflect staff responsibilities for fire safety and set in place appropriate means for recording and monitoring staff training.

All staff should receive induction training on or before their first day of employment. This may take the form of generic training. Were staff are working in areas where there is specific risks or hazards, the induction training must be supplemented by job-specific instruction as soon as their employment commences.

Training programmes should include, but not confined to, the following:-

- Basic fire safety;
- Good housekeeping;
- Action to take on discovering a fire;
- Procedures for evacuation;
- Staff responsibilities during a fire incident;
- Specialist roles.(switchboard staff, maintenance staff, fire marshals etc).

Maintenance Plan

There is a need to have in place an auditing and monitoring process reporting through the safety committee that ensures that all the items mentioned in the fire plan are adequately maintained. Many of the maintenance procedures are administrative. There is a tendency to automatically think of the engineering department when the subject of maintenance is mentioned. There are many maintenance duties that are the responsibility of administrators and medical staff.

Practical Management of Fire

Fire safety management is the practical skill of applying the policies and procedures so that in the event of fire there is a calm, professional and practised response to a fire incident that will occur.

When a fire occurs in any location if the staff are well trained a number of coordinated activities will start. A member of management who will be following agreed procedures to keep occupants safe and prevent fire spread will control all of those activities including confirmation that there is a fire to bring outside assistance from the Civil Defence. Once the fire is confirmed the following actions should commence:

Switchboard Operator

A team of Switchboard Operators when on duty has responsibility to make the emergency call when they receive information of a fire.

Fire Team Bleep Holders

There needs to be some form of communication system in place to ensure that fire team members are notified quickly of an emergency.

The fire team may consist of:-

- Maintenance Contract staff
- Administrative Personnel
- Clinical Staff
- Others

In case of fire, fire team members will respond to all fire calls and offer their specialist services to the Fire Team leader.

Emergency Telephone Numbers

The Hospital emergency telephone number is 9777 and is displayed on or near all telephones and on the fire action procedures.

Stages of Evacuation

Whenever possible, lateral or horizontal evacuation to a safe compartment is the preferred method of moving people away from a fire. This is achieved by:-

Stage 1 - Moving through one set of fire resisting doors and ensuring they are closed after people move through them to provide 30 minutes protection.

Stage 2 - Moving through two sets of fire resisting Doors (generally into another compartment and ensuring the doors are closed as people move through them, giving 60 minutes protection.

Managing Evacuation

Staff may be required to take wheel beds through the emergency route: therefore, the beds must be properly maintained to ensure free running wheels and ease of steering. Beds being used for emergency evacuation might be able to accommodate more than one patient (particularly useful with children), so speeding up the evacuation. Beds used for evacuation should be pushed clear of doors and parked with their brakes on where they will cause least obstruction. Smoke stop and Fire doors should be as each person passes through them to prevent the spread of smoke to unaffected areas.

Fire Prevention Measures

All textiles, furniture and furnishings supplied to the Royal Hospital should be clearly and durably labelled, or marked according to individual contract requirements and should conform to the requirements outlined in Royal Hospital policy

It is the responsibility of all managers and staff involved in procurement to ensure compliance when purchasing or contact the Fire Safety Supervisor on extension 5024 or 9511 for further information.

The hazards from burning furniture and textiles depend on the overall characteristics and environment. There is no truly flameproof material; the best that can be achieved is to increase the resistance to fire, thereby, reducing the fire hazard level. The term textiles and furnishings include the following:-

- Furniture
- Loose Covers
- Polypropylene Chairs
- Beds
- Divans
- Bedding
- Patients Sleepwear
- Textile Soft Floor Coverings
- Curtains, Drapes and Blinds
- Disposable Products
- Totally Soft Play or Physiotherapy Environments

Prevention of Fires caused by Smoking

The Royal Hospital currently has a 'No Smoking Policy' which manager's should refer to for further advice.

Prevention of Fires caused by Cooking

Cooking should only take place in properly constructed kitchens, which have the doors shut to prevent smoke and cooking fumes escaping. Cooking by whatever means should not be left unattended. All kitchens, where necessary, are equipped with one or more fire blankets and fire extinguishers.

Prevention of Fires caused by Faulty or Misused Electrical Items

Electrical equipment and wiring are a known cause of fires in Royal Hospital when misused or used in conjunction with faulty equipment, systems or bad practice

To help reduce the risk all electrical items used in Royal Hospital buildings are to be tested by a qualified electrician and if approved for use marked with a test label. A record of approved items is maintained in the Maintenance Department. Contracted companies should provide the test certificates prior to bringing equipment on site and annual testing thereafter.

Electrically Heated Appliances

Where electrically heated appliances, such as film processing units, incubator etc are liable to be kept switched on there should, in addition to the control thermostat, be a manual reset thermal cut-out fitted. It is the responsibility of the Line Manager to inform the Fire Safety Supervisor of such equipment.

Personal Electrical items

Flammable Substances Use and Storage

Flammable liquids in this policy refer to any liquid with a flashpoint of less than 55 C.

Many adhesives, polishes, paints and thinners used by the Engineering Department contain solvents of this nature and many liquids used in medical departments, laboratories and RDU's fall within this definition.

All flammable liquids pose a fire/explosion risk and must be stored safely. The quantities of any flammable liquid kept in departments ready for immediate use should be as small as possible having regards to the process or operations carried out. Ward /Departmental Managers are to ensure the minimum amount of flammable liquid is kept in a workplace and only then when in appropriate approved containers. Staff are to be made aware of rules, regulations and procedures for the safe handling, use and disposal of flammable liquids. Flammable liquids must never be used for any purpose other than that for which the Royal Hospital acquired them.

Highly flammable liquids of up to 50 litres maybe stored in a work area, provided they are kept in a metal fire resisting structure. Larger quantities must be stored in either bulk tanks or portable cylinders kept in open shaded compounds.

For less than 500cc of highly flammable liquid, provided the vessel is small closed and suitable for its purpose. No additional requirements are necessary for fire prevention purposes.

Flammable liquid stores should be clearly and boldly marked to indicate the flammability of the contents. But the safety warning signs and notices should be used as appropriate and posted in prominent positions, in particular bold type "No Smoking" and "No Naked Flame" notices should be displayed.

All flammable liquid containers should themselves be labelled.

The foregoing points should also be recorded in the Ward or Departments Risk Assessment as applicable.

Contractors

The risk of fire is increased when Contractor are on site. Outside Contractors present a greater fire risk than Royal Hospital staff because they are not as familiar with the premises as people permanently employed there. They cannot be expected to appreciate the fire risks, the necessary precautions and what action to take in the event of fire.

Contractors undertake work, which may be more hazardous than that normally, carried out on the hospital.

The site activity of all Contractors must be strictly supervised and controlled; even small works have the potential to cause serious fires. Management staff associated with any works must submit appropriate safety documentation indicating that the work will present no fire or safety risk to the building.

The safety documentation will include the use of Permit to Works and Hot Work Permits systems for those contractors and engineering technicians who by the nature of their work employ hazardous methods or processes in the course of their work. Only authorised engineers known as authorised persons (AP) from Royal Hospital Engineering Dep't may issue "permits to work" and the master records will be maintained within the Engineering Department.

Portable fire fighting equipment

Portable fire fighting equipment will be provided at strategic locations and indicated by appropriate signage. The equipment provided at these locations will be suitable for the risk in the immediate vicinity

All extinguishers are serviced and inspected annually, additionally they are discharged tested every five years. Suitably qualified contractors will be carried out these inspections and tests to current standards and the results will be recorded on each item of equipment. Test certificates are required from the servicing contractors and are retained by the Fire Officer.

Fire Blankets are positioned in those Kitchens and Restaurants where food is cooked and there is a possibility of persons being involved in a fire. Fire Blankets will be inspected annually, and the results will be recorded on each fire blanket carrier. The Fire Officer will retain a master record of service of fire blankets.

Fixed Fire Fighting Equipment

Fire Hydrants and by-pass arrangement connected to the water main as shown on the Fire sector should be tested 6 monthlies (fire hydrants are generally located in areas adjacent to risk and are indicated by fire hydrant marker posts)

a. Hose Reels

Hose Reels are fitted at each floor level in the premises will be serviced and inspected 6 monthly and annually by qualified contractors.

a. Dry Riser:

Dry riser fitted at each side around the Premises to support Civil Defence to connect their water hose.

Addressable/conventional Fire Alarm Systems

The Engineering Department and/or professional contractors will subject the Addressable/conventional fire alarm system operating throughout the hospital/clinic to a rigorous testing regime. The planned maintenance programme will ensure, zone, sector detector, call point and alarm siren is tested in compliance with the installation guidance and Royal Hospital policy.

Fire Alarms

General Response

If you hear the alarm **continuous ringing** you are in the zone where the fire has been discovered.

Action (Patient Access Area)

1. Ensure patients are moved from immediate danger, evacuate as necessary to the nearest designated internal assembly point.
2. Close doors.
3. Take correct extinguisher and tackle the fire if safe to do so.

More detailed instruction is contained within the local evacuation plan and procedures for each ward/dep't.

Hospital Local Evacuation Plans Patient Access Areas

Means of Escape

Hospitals are specially designed and constructed and resist the spread of fire.

Hospitals are designed as a series of compartments (Boxes) joined together by fire exits and wide corridors to ensure that patients can be quickly and safely moved around the hospital. In addition, there are sophisticated warning systems that ensure that fire is quickly detected, and the location known, so that a speedy response by fire terms will ensure that the fire is put out whilst small.

The design of hospitals makes certain that wherever you are in hospital, in the event of fire you can turn your back on, or move around, the fire and go to a place of safety within the building.

The evacuation principle for patient areas of the hospital is **Progressive Horizontal Evacuation (PHE)**, which is a three stage evacuation plan designed to move the minimum number of patients the shortest distance possible to a place of safety on the same floor and inside the hospital or outside the hospital as shown in the table below:

Main Hospital

Place of safety (Assembly Point) inside the Hospital	Place of Safety (Assembly Point) Outside the Hospital

• Maternity 1 to Maternity 2 • Maternity 3 to Paediatric 1 • Maternity 4 to GYN Oncology • SCBU to OB. OT and Anti Natal Clinic	<u>Level 0:</u> • Anti Natal Clinic • Laundry • Boiler House • Sub Station # 1 • Sub Station # 2 • Sub Station # 3
<u>Level 1:</u> • A&E Paediatric to OPD 2 • A&E Adult to Short Stay and Waiting Area • Male Medical 1 to Female Medical 2 • Paediatric 2 to Female Medical 1 • Paediatric 3 to Paediatric 4 • NICU / Haematology • Female Medical 2	<u>Level 1:</u> • Administration • Medical Records • Pharmacy • Radiography • OPD 1 • OPD2 • OPD Paediatric • Rehabilitation • CSSD • Engineering Department • General / Medical / Engineering Stores • Kitchen and both Cafeterias • Catering Department • Transport & Mortuary • Medical Gasses and all Plant rooms • Yard & Workshops
<u>Level 2:</u> • Operation Theatre to Recoveryroom, ICU, Intervention Lab 3 and OB OT • I CU Adult to Liver Room and Operation Theatre • Nephrology to Urology • Male Surgical to Female Surgical • Multi-Specialty Ward to Female Surgical • V I P Ward to Male Surgical	<u>Level 2:</u> • Biochemistry • Haematology • Histopathology • PGMC • Computer Departments • All Plant Rooms

Hospital Annex Buildings (Assembly Point) Outside the Buildings.

- Hyperbaric
- Genetics
- Diabetics
- MIC
- Oncology level 0
- Heart Centre Level 0

Risk Management

Basic rules for the reducing the fire risks in departments

The fire marshal is responsible for managing and making sure that staff are aware of the following: -

1. Making sure that there is no accumulation of rubbish, waste paper or other materials, which could catch fire in any room or corridors.
2. Ensuring that the means of escape are kept clear and un-obstruction at all time.
3. Making sure that flammable materials are kept in a metal cupboard when not in use.
4. Sufficient flammable liquid for daily use only should be kept in the department in a safe container suitably labelled.
5. Making sure that electrical wiring, plugs and sockets are regularly checked, that there is no overloading of sockets.
6. Only electrical equipment's approved by the engineering section should be allowed in the department.
7. Turn off electrical equipment when not in use (unless it is designed to be permanently connected).
8. Ensuring that material, which could easily catch fire, is not left near any source of heat.
9. Making sure that machinery and office equipment is well ventilated and regularly cleaned.
10. Switch off extract fans before you leave the area.
11. Making sure the fire detection system and call points are not obstructed / blocked.
12. Do not wedge open fire doors.
13. Making sure the fire fighting equipments is available and working.

Fire Alarm and Detection System

Sounds

A fire alarm in this area is indicated by

BELLS

A continues sounding fire alarm bell indicates that there may be a fire in your ward. Be prepared to initiate the local evacuation procedure if told to do so by the ward Marshal.

Testing

The fire alarm system is tested every

SUNDAY

The alarm will be tested for a maximum of 45 seconds only. Staff should be aware it is taking place during the alarm test reassurance of patients and visitors are needed. Any defect e. g. sounder not working is to be reported immediately.

Call Points

Manual break glass alarm call points are located so that staff have to work no more than 30 m and are generally located at exit points from wards/ departments and at the final exits from the building.

All staff should identify the call point in their place of work.

Escape Routes

Assembly Points

The fire marshal and the Fire Safety Supervisor will decide the assembly point. It will always be the next nearest to the department concerned.

Fire Safety Training for Royal Hospital Premises

Training responsibilities in Fire Safety

Under Omani Occupational Health and Safety Legislation there is a requirement on the private sector to ensure that all staff have suitable and sufficient information, instruction and training on call matters relating to safe working conditions and practices, and especially in emergency procedures of which fire is such a procedure.

It must be ensured that all staff and especially management staff know what to do in any emergency and that they are aware of the content of and status of the fire action plans and Fire Safety standards and where applicable the results of Fire risk assessments.

Also, when new staff starts work they should have induction training on emergency procedures.

The date and time and who gave training and who attended training must be recorded at all times. Any who attend late should be returned to work and a date rescheduled to ensure that no important training content is missed. This will ensure that you have an accurate record of training given and who attended.

Fire safety training content

Fire Safety training should be structured to be a logical progression and expansion of fire safety knowledge and awareness of staff.

The following subjects should be included: -

1. Fire development
2. Fire spread
3. Basic means of escape
4. The action to take on discovering a fire.
5. How to raise the alarm and what happens then.
6. The action to take on hearing the fire alarm.
7. The procedures for supervising the evacuation of visitors from the premises.
8. The arrangements for calling the Civil Defense.
9. Evacuation procedures for everyone to reach the assembly point at a safe place.
10. How to conduct fire drills.
11. The assembly points.
12. The location of firefighting equipment's.
13. Training in the use of fire-fighting equipment.
14. The location of the escape routes, especially those not in regular use.
15. How to open all escape doors including the use of any emergency fastening e.g. break glass bolts etc.
16. The importance of keeping fire doors closed to prevent the spread of fire, heat and smoke
17. How to isolate machines, equipment etc as necessary.
18. How to isolate power supply sources.
19. How to isolate piped medical gases.
20. Why not to use lifts (if provided)
21. Evacuation of patients, visitors or staff.
22. Importance of general fire safety and good housekeeping.

25. The need to report equipment faults and malfunctions.

26. How to deal with spillages

Fire Drills

A practice fire drill should be carried out at least once every twelve months in each ward/department or building. For those areas of high risk, areas with complex procedures or large numbers of people fire drills should be carried out more frequently.

Fire escape routes should always be available for use in the building. But it must be remembered that people are creatures of habit and will tend to leave the building by the same route they entered the building.

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ROYAL HOSPITAL

FIRE SAFETY

Information Handbook



It is hoped that the information contained in this handbook will provide the reader with sufficient information to be able to:

- Gain a basic knowledge of the chemistry of combustion.
- Select the appropriate equipment to extinguish a fire.
- Operate the hospital firefighting equipment.
- Identify and deal with electrical hazards.
- Recognize the dangers of smoking and be aware of the “No Smoking” policy.
- Discuss basic fire prevention methods.
- Appreciate the extreme dangers of “Toxic Smoke” in a fire situation.
- Demonstrate a sound knowledge of the hospital fire reporting procedures.
- Take appropriate action in the event of a fire.

FIRE SAFETY

Fire destroys life, property and jobs, and, in addition, causes damage to the environment.

Everyone here at the Royal Hospital is responsible for fire safety. All employees should have sufficient knowledge so that they can identify a potential fire and take the correct action in order to prevent it happening. Each employee must be proficient with the hospital fire reporting procedures and also be capable of extinguishing any fire in its early stages using the firefighting equipment provided.

With fire, as with other unwanted happenings, prevention is clearly better than cure. A basic knowledge of combustion, an understanding of fire prevention techniques and a good degree of competency in the use of the hospital fire equipment will go far in preventing a catastrophe from happening.

THE CHEMISTRY OF FIRE

are generated.

The products of combustion

Heat, light, smoke and asphyxiating and toxic gases are produced by fire. In a very hot, well ventilated fire, combustion is complete. All the carbon is converted to carbon dioxide, all the hydrogen is converted to steam. Oxides of various other elements such as Sulphur and nitrogen are produced.

Transmission of Heat

The transmission of heat is responsible for the start and continuation of most fires. Heat is transferred by one or more of the following methods.

Conduction – transmission of heat by contact

Eg. A cold spoon placed in a hot liquid will absorb heat and in turn transfer that heat to the hand that holds it.

Convection – transportation of heat by movement of heated liquids or gases.

Eg. Heated air expands and rises. This raises the temperature in areas to which it travels.

Radiation – transmission of heat from one body to another by heat rays passing directly across the space between them. Radiated heat normally travels in straight lines but can be reflected or magnified.

Eg. Open bar heater and sunlight.

The fire triangle

Fire has three essential components: **fuel, oxygen and heat** - forming the so-called fire triangle.

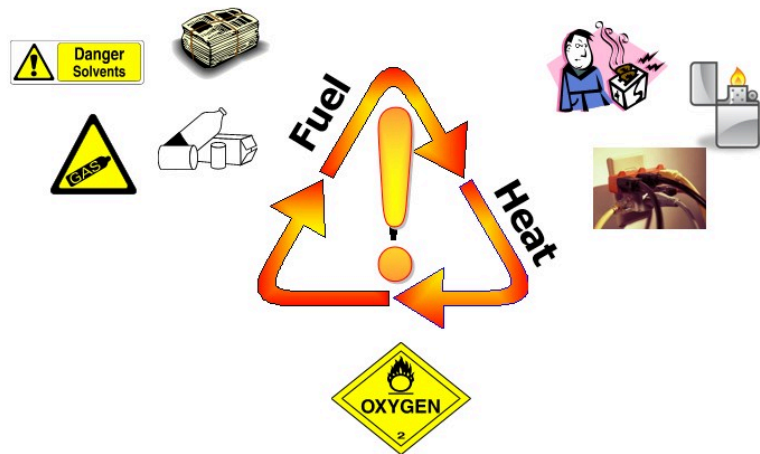
Fire occurs where these components are brought together in the right conditions. If one or more of the components of the fire triangle is removed, the triangle collapses and the fire will be extinguished. This can be done by:

- Cooling the fire to remove the heat, usually with water.
- Starving the fire of fuel.
- Smothering the fire by limiting its oxygen supply.

One example of smothering a fire would be to cover a burning oil pan with a dampened cloth or fire blanket. The fire will consume all of the oxygen in the cavity between the surface of the burning oil and the damp cloth or fire blanket and self-extinguish. Another means is to drive away the oxygen-containing air from the vicinity of the fuel and to replace it with carbon dioxide or other gases which will not support combustion.

Another technique is to apply a dry chemical powder in the form of a cloud. This extinguishes the fire by interfering with the chemical reactions of the flame.

Fuel	Starvation
Heat	Cooling
Oxygen	Smothering



The Classifications of fire

International specifications classify fire according to the nature of the fuel. The most suitable extinguishing agents for a particular type of fire can then be related to the classification.

Class 'A' Fire

Solid carbonaceous materials such as paper and wood etc. This is the most common type of fire and the most effective extinguishing agent is generally water.

Class 'B' Fire

Flammable Liquids, normally petroleum spirit based. Foam, CO₂, and dry powder are normally used to extinguish these fires.

Class 'C' Fire

Flammable gases, mostly liquefied petroleum Gas (LPG), but may also include Liquefied Natural Gas (LNG). At the basic level, other than calling the emergency services and evacuating a building, starvation is the only method that should be attempted when dealing with this type of fire.

Fat and cooking oil fires

Electrical fires are not considered as a class of fire but as a separate group. Electricity does not catch fire but is normally a means of ignition for fuels. Any fire involving electricity must be either an A, B, C, D or F class fire. The procedure is to cut off the supply of electricity and use the extinguishing appropriate to whatever is burning. In cases where this is not possible a nonconductive agent such as general-purpose dry powder, or carbon dioxide must be used.

FIRE EXTINGUISHERS

It must be noted that in all circumstances, a fire extinguisher is designed and installed to be used only as a first aid fire appliance. A Fire Extinguisher is not intended to be used for a prolonged attack at a fire. Any attempt to extinguish a fire must only be made if it is safe to do so and where an escape route is clear at all times.

There are two types / designs of fire extinguisher in use at the Royal Hospital. These are stored pressure, where the extinguisher is pre-charged and has a small gas contents gage at the neck. The second is gas cartridge operated where the extinguisher is fitted with an internal gas charge. This type of extinguisher has no contents pressure gage.

In both designs of extinguisher there are 3 types of extinguishing agent available. These are, Multi-Purpose Chemical Dry Powder, Foam and CO₂.

Characteristics of Fire Extinguishers



Dry Powder Fire Extinguishers are coloured blue or red with a light blue band. They are multi-purpose and can be used to fight Solid, Liquid and Gas fires. They are also safe to use at fires involving energised electrical appliances. They are generally in the region of 6 to 9 Kg in weight. Because of their multi-purpose capability, where ever they are provided they should always be the first choice of extinguisher taken from any fire point. Whilst dry powder fire extinguishers are non toxic, a large residue of very fine powder will result and the area should be thoroughly ventilated immediately after use.



Foam Fire Extinguishers are coloured cream, or red with a cream band. The foam is normally applied via a short hose and a spray nozzle. They can be used to fight Solid and Liquid fires. They are not recommended for use at fires involving energised electrical appliances. Their capacity is normally approximately 5 to 6 litres, so they can be slightly heavy. Operation is easy, with the operator sweeping the foam across the surface of the fire for an effective knockdown.



CO2 Fire Extinguishers are coloured black, or red with a black band. These are quite heavy. Discharge is normally very noisy and they must be handled carefully to avoid dry ice burns. They can be used to fight B and C class fires and are also safe to use at fires involving energised electrical appliances. Unlike dry powder extinguishers which are very messy, the CO2 leaves little or no residue so is the choice around delicate computer and biomedical equipment. Ventilation should take place following discharge.



Fire blanket is a highly flame resistance that can be used to extinguish a small fire or to wrap around a person in case of fire. Fire blanket made from 2 layers of woven glass fiber fabric and an inner layer of fire retardent film

Operation

Guidelines for operation of all fire extinguishers are as follows :

- Remove extinguisher from cabinet or bracket at the fire point.
- Take extinguisher to the fire.
- Remove safety pin from the discharge mechanism.
- Aim hose / nozzle at the seat or heart of the fire.
- Stand back two to three metres.
- Discharge contents by squeezing the trigger or striking the cap.

Maintenance and servicing

Fire extinguishers are inspected and maintained by a certified contractor on an annual rotational basis at the Royal Hospital. Between these inspections the user department can also inspect the extinguisher in order that it remains in a fully serviceable condition at all times.

A simple visual inspection by the user department should be carried out weekly as follows:

- Check fire extinguisher and bracket remain firmly fixed to wall or that the fire extinguisher is secure in the fire cabinet
- If the extinguisher has a gauge check indicator needle is in the green section.
- Check the safety pin is present and in it's correct position with the wire or plastic seal unbroken..
- Check the hose or nozzle for signs of leakage or discharge of contents.

- Call fire safety department immediately if any defects are found.

INSTALLED FIRE PROTECTION EQUIPMENT (The Royal Hospital Complex)

In addition to the portable fire extinguishers there are also installed fire protection systems. These are fixed water hose reels, automatic and manual fire alarm and detection systems, fire escape stairs, fire resistant doors between fire compartments and emergency fire escape doors. Fire points are provided where possible, grouping fire equipment together at a single instantly recognisable strategic point.

Fire hose reels (water)



Fire hose reels are an installed device comprising a rubber hose of between 20 to 40 metres in length attached to a red drum. This drum is fixed to the wall. A dual action spray / jet nozzle is attached for discharge of water.

The hose will pressurize automatically as the hose is run from the drum. Aim the nozzle at the base of the fire and twist the nozzle. Water will flow continuously under pressure. To stop the flow of water simply twist the nozzle in the opposite direction. The danger of mixing water and live electricity should never be overlooked. An initial attack at any fire should always be made using a portable fire extinguisher, and a hose reel should only ever be used as a very last resort.

Automatic fire detection & alarm system

Automatic fire detection & alarm systems are installed at the hospital complex. This system comprises automatic smoke & heat detectors linked to the main fire alarm panel situated at the hospital switchboard.

Depending on the type of detector it will detect the presence of smoke or an increase in temperature. An alarm will sound in the vicinity of the detector and the location of the detection will be reported to the main alarm panel. To ensure their serviceability the detectors are tested routinely by the Maintenance Department on a rotation basis. Testing is carried out each Sunday approximately 12.00 hrs.



The manual fire alarm system at the Royal Hospital comprises a red box fitted to the wall, often referred to as a “Manual Call Point”. This is a standard “Break Glass” system. When the glass is broken an audible alarm will sound in the vicinity, and the location of the call will be reported to the main alarm panel.

Fire resistant doors



Fire resistant doors are installed throughout the hospital. These have typical fire resistance ratings of 30 and 60 minutes protection depending upon the location to which they are fitted. These doors are glazed with re-inforced Georgian wired glass and are fitted to double and single leaf frames. Fire doors act as fire and smoke protected points of entry and exit to fire resistant compartments throughout the hospital.



Unless the fire door has electronic retainers linked to the fire alarm system the door should remain closed at all times. They should never be wedged or propped open by heavy objects, neither should they be blocked. If a fire door is found to be blocked, wedged or propped open, or damaged in any way the fire safety department should be informed immediately.

Emergency exit doors

Emergency exit doors are installed to allow fast and direct access to safety in an emergency situation. These often pose a threat to security as they need to be accessible from inside the building at all times without being locked. This would obviously leave several points of uncontrolled entry through which unauthorised persons might exit or enter the buildings.



To overcome this problem where security control is required, selected emergency exit doors at the Royal Hospital are fitted with a panic device. This is a small device fitted to the door and frame with a panic button mechanism which allows the door to be opened easily in an emergency. When a door fitted with a

panic bolt is opened a very loud alarm will activate to alert security personnel.

Fire escapes



The Royal Hospital has fire protected stairwells. These fire protected stairwells are designed to be used as fire escapes. No part of these stairwells, the landings or doorways must ever be obstructed in any way. Any obstruction, must be reported to the fire safety department immediately.

Elevators

The elevators at the hospital are built in a fire protected shafts. Check unit local fire plans for instructions regarding the use of elevators during fire emergencies.

SMOKING AND THE “NO SMOKING” POLICY

ممنوع التدخين



NO SMOKING

Smoking and smoking materials are without doubt one of the major causes of fires.

Wherever smoking is taking place, ash trays should be provided. These should be of a non combustible material and not oversized. Waste bins should be made of non combustible material and have a close fitting lid.

The Royal Hospital has a no smoking policy. 'No Smoking' signs are posted throughout the complex.

Staff, patients and visitors are required not to smoke and should at all times respect the no smoking signs. However, the signs are at times either missed or ignored. When this occurs it is the duty of all staff to point out the hospital 'No Smoking' policy in order to minimise the dangers from smoking when ever possible. Staff should never become embroiled in a heated debate with patients or others regarding no smoking, If such a situation escalates the department senior staff must be informed immediately and the situation resolved. If the situation can not be resolved it should be referred immediately to Security.

Smoking and an Oxygen enriched atmosphere.



Staff should always be alert and watch for patients smoking or, bedside smoking by visitors, particularly where medical oxygen is present. Smoking in the presence of oxygen is extremely dangerous. Pure oxygen promotes rapid combustion. Plastic items such as combs, spectacle frames, ball pens etc., which might normally burn sluggishly or melt, are greatly assisted in the presence of oxygen and burn fiercely. Oxygen can also permeate a patients bedding which, if ignited will burn very aggressively and is extremely difficult to bring under control.

ELECTRICITY AND ELECTRICAL SAFETY

Electricity and electrical appliances are another major cause of fires. If you identify an electrical hazard in your workplace, report it immediately to Maintenance Control.

SOME COMMON ELECTRICAL FIRE HAZARDS ARE:

Electrical cables and extension leads

Any electrical current passing through a cable produces heat, and the higher the wattage of the appliance being used the higher will be the temperature. External damage or wear could mean that the lead has been damaged internally in such a way as to allow sparking between the two conductor wires; the temperature of these sparks can easily reach 10,000 degrees

Electrical plugs

2-Pin plugs including 2-pin laptop chargers and 2-pin mobile phone chargers are not permitted to be used in the Royal Hospital. These are dangerous, incompatible with the hospital electrical supply and need to be adapted in order to be used.



Electrical plugs should be 13 amp three pin plugs of good quality, selected to fit directly into the socket without the need for an adaptor. A fuse of the correct rating appropriate to the appliance should be fitted into the plug, and there should be no movement at the point where the wires enter the plug. Loose wires and bad connections to terminals may give rise to arcing, with the production of very high temperature sparks. Only one appliance should ever be connected to one plug. An appliance should never be powered by connecting bare wires to the socket.

Electrical adaptors



Apart from the danger of overloading a circuit the sheer weight of adaptors and plugs can cause the adaptor to partially withdraw from the wall socket outlet giving rise to poor electrical contact, which in turn will give rise to arcing with the production of high temperature sparks. A plug suitably matched to an electrical outlet socket should always be fitted to the electrical appliance. This will eliminate the need for an electrical adaptor. Where multiple sockets are absolutely necessary the Engineering department should be contacted for advice.

Fluorescent lights

before the fire is detected. Flickering fluorescent lights should be reported to the maintenance department as soon as possible.

Electrical fuses

When a fault develops in an appliance and it starts to draw too much current the fuse should melt (safely) and thereby cut off the current and protect the appliance. If a fuse of too high a rating is in place, the appliance may overheat to the extent that it catches fire before the too highly rated fuse will melt.

Faulty electrical equipment

Faulty electrical equipment in the workplace must be dealt with swiftly to prevent accidents and fires. If electrical equipment fails for any reason, the following procedures need to be followed:

- Switch off appliance immediately.
- Remove plug from wall socket.
- Place an “Out of Service” label over the equipment on / off switch.
- Inform dept. supervisor and raise work order with Maintenance Control (ext 9512).

GENERAL FIRE PREVENTION

Despite all our best efforts to prevent fires from starting, fires will continue to occur and therefore, apart from the obvious fire prevention methods previously discussed under the headings smoking and electricity, we must be able to identify two other types of hazards. Those which would allow the fire to spread through the building once it has started, and those that would put lives at risk in the event of a fire. In short, there are three categories of fire hazards that have to be considered:

- **Those that cause fires.**
- **Those that spread fires.**
- **Those that put lives at risk.**

In order to understand why the examples given below do in fact constitute a fire hazard you should be aware of three facts:

Most people who die in fires are killed by inhalation of toxic smoke and gas, and about 75% of fire spread within buildings is caused by the movement of the extremely hot air and gases,

Some of the principle causes of electrical fires are:

- Misuse of, or faults in, electrical equipment – including leads, plugs, and fuses,
- Careless disposal of cigarettes and other smoking materials,
- Hot work – welding, grinding and cutting etc – undertaken by contractors.

Some examples of each of the three identified categories of fire hazards are described below.

- Damaged and worn electrical leads.
- Misuse of electrical adaptors.
- Misuse of electrical extension leads and using poor quality extension leads.
- Flickering fluorescent lights.
- Poor/loose connections inside electrical plugs.
- Using an electrical fuse of too high a rating.
- Combustible materials in contact with, or too close to, sources of heat. □ Lighted cigarettes left on edges of tables or desks, or carelessly discarded.

HAZARDS THAT ALLOW FIRES TO SPREAD WITHIN A BUILDING

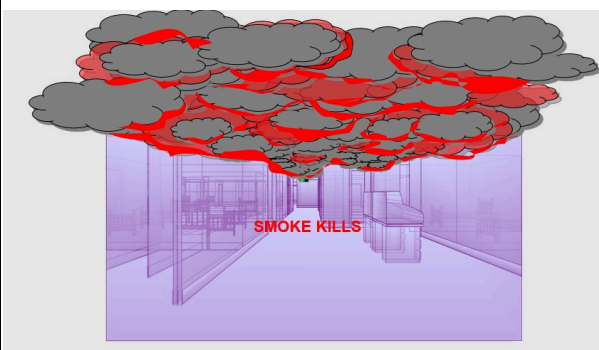
- Fire doors left open.
- Broken or missing tiles in suspended ceilings.
- Large undivided ceiling voids
- Unsealed cavities around cables, conduit and ducting passing between walls dividing rooms, offices and fire compartments.

HAZARDS THAT PUT LIVES AT RISK IN THE EVENT OF A FIRE

- Open fire doors that lead onto escape routes.
- Obstructed escape routes.
- Hidden fire extinguishers.
- Obscured or missing fire exit signs.

Fire prevention (in basic terms) is about common sense, observation and communication. The various fire hazards described above are those which are frequently encountered when carrying out routine inspections. By identifying, and then removing, rectifying, or reporting the hazards, you can play a vital part in making your department a safer place in which to work.

SMOKE & TOXIC GASSES



property damage.

Under certain conditions materials will burn to produce smoke and particulate matter which can be carried through ducts and ventilation shafts to locations far away from the immediate area of the fire. This endangers others.

The contents and structure of buildings often contain large quantities of plastics, for example polyvinylchloride, polyethylene and polystyrene, with a variety of additives. When these plastics are involved in a fire, even at low temperature, dense smoke and corrosive combustion gases are produced very quickly, which depending on the nature of the polymer may include: Hydrogen chloride, chlorine, sulphur dioxide, oxides of nitrogen and also ammonia. This already toxic atmosphere is further enhanced by water vapour produced in the combustion process and normal humidity, resulting in the gases dissolving in water to produce further corrosives such as hydrochloric acid.

The above lethal cocktail of gases and corrosives is very common. Eyes, sinuses and the respiratory system are attacked making survival in such an environment impossible and the chance of escape extremely difficult. Smoke and heated gases rise. If there is no alternative but to make an exit to safety through an area filled with smoke, the victim should travel as low or as close to the floor as possible where a supply of fresh air may be found.

The Golden Rule in a smoke filled environment is :

GET DOWN LOW AND GO, GO, GO !!!

It must be realised that the confinement of smoke in a fire situation is as equally important as confining the fire itself. Remember to close all doors and windows and, where necessary, switch off the air conditioning as you evacuate.

ACTION TO TAKE IN THE EVENT OF A FIRE HOSPITAL FIRE REPORTING PROCEDURES AND

- A fire could occur at any time of the day or night. If you discover a fire in your department follow these steps :
- Shout “FIRE” Summon the assistance of others in your area.
- Remove all patients from immediate danger.
- Raise the alarm by breaking the glass at the nearest fire alarm call point.
- Dial the Fire Emergency Number (Ext. 9777) and notify the operator of the exact location of the fire and nature of the fire..
- If it is safe to do so, attempt to extinguish the fire.
- If the fire is out of control, prepare to evacuate. It is important to remember to close all doors as you leave and follow the instructions of your wards local fire plan.

The previous steps give only basic guidelines to be followed in the event of discovering a fire. However, only the circumstances at the time of any fire emergency will dictate exactly what action you must take. Everyone dealing with a fire should be prepared to make an immediate analysis of the situation. Delegation of tasks and a calm command of the situation is your job until relieved by a higher authority.

Some factors for your consideration are :

- The location of the fire in respect of the nearest fire alarm, nearest telephone and nearest fire fighting equipment.
- In patient areas, the location of the patients, the number of patients, their ages and the ambulatory condition and their proximity to the fire.
- The size and state of the fire and it's effect on your direction of travel to the nearest exit or safe area.
- The number of staff available to assist due to the time of year or the time of day etc..

Evacuation

Evacuation is movement, either horizontal or vertical from a dangerous or potentially dangerous area to one of safety.

There are four stages of evacuation. Each may be a separate and complete operation or all four may have to be used in successive stages if circumstances so require.

Partial Evacuation

This is removing one or more patients from immediate danger to the nearest place of safety in a fire situation. When patients are removed, an attempt must be made to extinguish the fire with extinguishers. If this is impossible, the door must be closed to contain the fire. If the fire continues to grow and threatens to break through, then the next step is to proceed with progressive horizontal evacuation.

Progressive Horizontal Evacuation

This evacuation takes place when fire or heavy smoke from a room threatens to spread to adjoining areas. All patients should be moved as soon as possible. They should be walked to safety, or removed on wheeled beds, wheelchairs, or stretchers led by someone to the area predesignated in the ward recovery plan remote from the fire and on the same floor level. All rooms must be checked for remaining occupants and all doors closed. When horizontal evacuation is ordered, the personnel in the pre-designated safe area should assist in the removal. When all persons have been moved to the safety of the safe area all fire doors should be closed and sealed. Finally, remain alert for the need of any further movement. As you move it is important for you to stop and feel for heat before opening any door. Any heat felt through a door indicates that there may be a fire on the other side and that the door should not be opened unless absolutely necessary.

Vertical Evacuation

This is the downward movement of patients and casualties to a safe area on the level below. Movement to lower levels can be made via the service ramp. It is normal practice not to use the elevators in a fire emergency but with the completion of the ONCC extension here at the Royal Hospital the elevators there may be used to move from the main hospital to lower levels in a fire situation as they are not connected to the main hospital services. In reverse this also allows for downward movement of the ONCC extension via the main hospital elevators.

The stairwells at the Royal Hospital are enclosed fire compartments and offer a place of safety during vertical evacuation of ambulatory persons. Remember, the extraction of non ambulatory patients in a vertical downward direction using stairs is traumatic for patients, and extremely exhausting for rescuers. All other possibilities and options should be explored before making a decision to use the stairs with non ambulatory patients.

Total Evacuation

This means vacating all the floors to a safe place. A 'Safe Place' will in most cases will mean leaving the building and relocating to an alternative Health Care Facilities. This is an enormous operation and requires the help of everyone available. This action must be undertaken floor by floor with enough trained help above and below to keep traffic moving quickly and properly. If this ever became the case a full pre-designated recovery plan for all services would need to be initiated immediately to preserve life.

Internal (Fire) Evacuation Planning



The Fire Safety Department issues fire plans for all departments. With the co-operation and expertise of individual departments staff and management these basic instructions can be further developed. In the hospital environment there can be no generalising and therefore each of the plans will be tailored to the requirements of the particular department taking into consideration its own hazards and risks etc..

Each fire plan will be accompanied by a powerpoint presentation issued which can be used on the unit for periodic table top training. This will provide an opportunity for constant re-orientation and fire safety awareness of staff to their own workplace and will also provide an opportunity to discuss and further develop the plans.

Assembly point



Assembly point sign is a place where people should go in the event of an emergency. The location of assembly point should be in a safe area out side of the building

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